

SRI SATYA VENKATA VINAY MAHESH PATNALA

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PROFESSIONAL SUMMARY

Robotics and Electronics Engineer with an MSc from Heriot-Watt University and hands-on experience across the full robotics stack from embedded hardware and ROS to computer vision and machine learning. Built and deployed projects in autonomous manipulation, human-robot interaction, and AI-powered defect detection using industry-relevant tools including Python, PyTorch, COLMAP, and Arduino. Member of the Edinburgh Centre for Robotics and Heriot-Watt Maker Society. Seeking graduate roles or internships in robotics engineering, computer vision, ML engineering in the UK.

EDUCATION

MSc Robotics

Jan 2025 – May 2026

Heriot-Watt University, Edinburgh | Edinburgh Centre for Robotics

- Dissertation: Defect Detection in Complex Environments using NeRF & 3D Gaussian Splatting supervised by Dr. Maria Koskinopoulou & Dr. Abderrahim Halimi.
- Modules: Robotic Mechanical Systems, Machine Vision, Robotic System Science, Intelligent Robotics, Human-Robot Interaction.

B. Tech Electronics & Communication Engineering

June 2018 – July 2022

Vel Tech Rangarajan Dr. Sagunthala R&D Institute, Chennai, India

- Published: “Design of Printed Antenna for Next Generation Wi-Fi Devices” -International Conference on Emerging Technologies, Chennai, May 2022.
- Focus on embedded systems, Arduino, AI, and hardware prototyping.

TECHNICAL SKILLS

3D Reconstruction & Vision: NeRF, 3D Gaussian Splatting, COLMAP, Nerfstudio 1.1.5, Point Cloud Processing, Chamfer Distance, OpenCV

Deep Learning & AI: PyTorch, CNN Architectures, Ollama LLM, CUDA GPU Programming, Google Colab

Robotics & Simulation: ROS, Webots, Robot Kinematics, HRI Design

Programming & Tools: Python, C, MATLAB, Bash/Linux, ffmpeg, Git, VS Code

Embedded & Hardware: Arduino, ESP32-C3, FreeRTOS, ESP-IDF

KEY ACADEMIC PROJECTS

Defect Detection via NeRF & Gaussian Splatting - MSc Dissertation

Supervised by Dr. Maria Koskinopoulou & Dr. Abderrahim Halimi

- Built an end-to-end pipeline: raw video → ffmpeg extraction → COLMAP pose estimation → Nerfstudio training (Nerfacto, Splatfacto) → point cloud export → defect analysis.
- Implementing and benchmarking two neural rendering approaches **Nerfacto** (volumetric NeRF), **3D Gaussian Splatting (Splatfacto)** evaluated on PSNR, SSIM, LPIPS and real-time FPS metrics.
- Preprocessing raw video frames using ffmpeg for frame extraction and filtering, applying image quality checks to ensure optimal input for COLMAP pose estimation and consistent model training.
- Developed parallel 2D and 3D defect detection strategies; the 2D pipeline employs illumination-normalised heatmaps and difference maps for surface anomaly visualisation, while the 3D pipeline applies Chamfer Distance-based point cloud comparison between clean and defective Splatfacto reconstructions to quantify geometric deviations.

4-DOF Autonomous Robotic Arm - Robotic Mechanical Systems

Supervised by Dr. Xianwen Kong

- Designed and 3D-printed a 4-DOF robotic arm; implemented full forward/inverse kinematics pipeline in MATLAB.
- Integrated OpenCV-based vision system for object detection and localisation, enabling closed-loop pick-and-place operations with Arduino actuation.

CheerMate: HRI Fitness Coach - Human-Robot Interaction.

Supervised by Dr. Christian Dondrup

- Programmed a Marty V2 robot as an interactive fitness coach using Python and Ollama LLM for natural language dialogue.
- Developed multimodal feedback system (speech + physical movement) and QR-based activity tracking for personalised coaching sessions.

ESP32-C3 Embedded Waveform Generator - Embedded Software

Supervised by Dr. Kieran O'Leary

- Bare-metal app generating two configurable periodic signals (DATA & SYNC) via Arduino IDE and ESP-IDF (FreeRTOS).
- Surname-derived timing (T_ON 1100–1600 μ s, 11 pulses, 6500 μ s idle); all parameters verified on oscilloscope.

PROFESSIONAL EXPERIENCE

Python & Data Science Intern

Feb 2024 – April 2024

El Systems Services, India

- Completed comprehensive training on Python with a focus on Data Science, acquiring skills in data manipulation, analysis, and visualization.
- Utilized machine learning algorithms to predict insurance claims, enhancing the decision-making process.
- Developed skills in coding, data analysis, cybersecurity, and forensic technology principles

OTHER EXPERIENCE

Customer Service Assistant

April 2025 – Present

Pasta Evangelists, Edinburgh, UK

- Maintained high operational standards in a fast-paced environment while completing full-time MSc studies.
- Collaborate effectively within a diverse team to ensure efficient workflow and adherence to strict health and safety regulations.
- Demonstrate reliability, time management, and the ability to work under pressure to meet production targets.

PUBLICATIONS & ACHIEVEMENTS

- Conference paper: “Design of Printed Antenna for Next Generation Wi-Fi Devices” -International Conference on Emerging Technologies, Chennai, May 2022.
- Affiliated researcher, Edinburgh Centre for Robotics (ECR) -one of the UK’s leading robotics research hubs.
- Dissertation findings being prepared for potential journal submission on NeRF-based industrial inspection pipelines.

EXTRACURRICULAR & LEADERSHIP

- Active participant in weekly Makerspace sessions, collaborating with cross-disciplinary students to prototype and build hardware/software projects.
- Utilise society equipment including 3D printers, laser cutters, and resin printers to rapidly prototype components for robotics coursework and personal projects, complementing dissertation research.
- Engage in skill-building workshops on 3D modelling and Arduino, and contribute to inter-society collaborative projects, strengthening both technical and teamwork competencies beyond the classroom.